Document that contains a tracing and error log from Langfuse and a summary of the agent's performance including any issues you encountered and how the issues were resolved. In the summary, indicate whether RAG either 1) enhanced agent performance, 2) degraded agent performance, or 3) had no impact on agent performance, and provide an explanation for your assessment. If RAG had no impact or it degraded performance, suggest some ways you might improve RAG performance.

Anh, Nirvaan, Shital Build 4

Find the Langfuse tracing here

Ask <vague conceptual question> “what are the most important questions? answer them. what can we learn from this dataset and analysis to apply to daily life?”

```
> ask what are the most important questions? answer them. what can we learn from this dataset and analysis to apply to dai
ly life?

=== ROUTER DECISION ===
{
  "mode": "answer",
  "note": "This is a conceptual question that does not require running analysis. Important questions could include: How do
es daily screen time affect sleep quality? What is the relationship between stress levels and sleep duration? Analyzing th
ese factors can help individuals manage their screen time and stress for better sleep and overall well-being."
}

Note: This is a conceptual question that does not require running analysis. Important questions could include: How does da
ily screen time affect sleep quality? What is the relationship between stress levels and sleep duration? Analyzing these f
actors can help individuals manage their screen time and stress for better sleep and overall well-being.


ERROR: Router 'mode' must be 'tool' or 'codegen'. Try again.
```

- Correctly does not run as it identifies the question as conceptual.
    - So we need to more explicitly ask something as a question to be answered.

Ask <analysis question> “ generate code to analyse this data”

```
> ask generate code to analyse this data

=== ROUTER DECISION ===
{
  "mode": "codegen",
  "code_request": "Perform a basic profile analysis of the dataset, summarize numeric columns, and summarize categorical c
olumns for 'gender' and 'occupation'.",
  "note": "This code will provide an overview of the dataset, including descriptive statistics for numeric variables and f
requency counts for categorical variables."
}

Note: This code will provide an overview of the dataset, including descriptive statistics for numeric variables and freque
ncy counts for categorical variables.


=== RAG CONTEXT RETRIEVED FOR CODEGEN ===

[Reference 1 | source=analysis_guides/tool_selection_rules.md | section=Dataset overview requests | score=0.539]
If the user asks to describe or summarize the dataset, start with a basic profile and variable-type split.

---

[Reference 2 | source=analysis_guides/basic_data_analysis_workflow.md | section=Step 1: inspect the dataset | score=0.518]
Start by profiling the data and identifying variable types.

---

[Reference 3 | source=build0_tool_notes/basic_profile.md | section=Outputs | score=0.500]
A summary object or dictionary with information such as:
- number of rows
- number of columns
```

- Explicitly stating to analyze the dataset results in a router decision for codegen where rag can be used

- Human in the loop is also working, approving the code, as well as asking to run it.

Ask “what are the gender differences for sleep / phone use?”

```
> ask what are the gender differences for sleep / phone use?

=== ROUTER DECISION ===
{
  "mode": "codegen",
  "code_request": "Analyze the differences in sleep duration and phone usage before sleep between genders. Calculate the m
ean and standard deviation for both metrics, grouped by gender.",
  "note": "This request involves a comparative analysis that is not directly covered by a single tool."
}

Note: This request involves a comparative analysis that is not directly covered by a single tool.


=== RAG CONTEXT RETRIEVED FOR CODEGEN ===

[Reference 1 | source=analysis_guides/tool_selection_rules.md | section=Dataset overview requests | score=0.405]
If the user asks to describe or summarize the dataset, start with a basic profile and variable-type split.

---

[Reference 2 | source=analysis_guides/basic_data_analysis_workflow.md | section=Step 1: inspect the dataset | score=0.402]
Start by profiling the data and identifying variable types.

---

[Reference 3 | source=analysis_guides/basic_data_analysis_workflow.md | section=Step 3: summarize variables | score=0.374]
Use descriptive statistics for numeric variables and frequency summaries for categorical variables.

---

[Reference 4 | source=analysis_guides/basic_data_analysis_workflow.md | section=Step 5: examine relationships | score=0.36
6]
```

Ask “what are the most important questions? answer them. what can we learn from this dataset and analysis to apply to daily life?”

```
> ask what are the most important questions? answer them. what can we learn from this dataset and analysis to apply to dai
ly life?

=== ROUTER DECISION ===
{
  "mode": "answer",
  "note": "This is a conceptual question that does not require running analysis. Important questions could include: How do
es daily screen time affect sleep quality? What is the relationship between stress levels and sleep duration? Analyzing th
ese factors can help individuals manage their screen time and stress for better sleep and overall well-being."
}

Note: This is a conceptual question that does not require running analysis. Important questions could include: How does da
ily screen time affect sleep quality? What is the relationship between stress levels and sleep duration? Analyzing these f
actors can help individuals manage their screen time and stress for better sleep and overall well-being.


ERROR: Router 'mode' must be 'tool' or 'codegen'. Try again.
```

- I wanted a challenging question to see how RAG was used, but it identified it as a conceptual question.
    - Questions for our group: is this correctly identified? As a group do we want these kinds of broad questions to not result in codegen to try and answer them?

Ask “what are differences in sleep / phone usage for college students vs adults? which group is better, analyse what impacts each group, is anything diff?”

```
> ask what are differences in sleep / phone usage for college students vs adults? which group is better, analyse what impa
cts each group, is anything diff?

=== ROUTER DECISION ===
{
  "mode": "codegen",
  "code_request": "Analyze the differences in sleep duration and phone usage before sleep between college students and adu
lts. Create summary statistics for both groups, including means and standard deviations. Additionally, perform a multiple
linear regression for each group to see what factors impact sleep quality, including daily screen time, stress level, and
caffeine intake.",
  "note": "This request involves multiple analyses and comparisons that are not covered by a single tool."
}

Note: This request involves multiple analyses and comparisons that are not covered by a single tool.


=== RAG CONTEXT RETRIEVED FOR CODEGEN ===

[Reference 1 | source=analysis_guides/tool_selection_rules.md | section=Dataset overview requests | score=0.447]
If the user asks to describe or summarize the dataset, start with a basic profile and variable-type split.

---

[Reference 2 | source=analysis_guides/basic_data_analysis_workflow.md | section=Step 1: inspect the dataset | score=0.436]
Start by profiling the data and identifying variable types.

---

[Reference 3 | source=analysis_guides/basic_data_analysis_workflow.md | section=Step 3: summarize variables | score=0.387]
Use descriptive statistics for numeric variables and frequency summaries for categorical variables.
```

- In an effort to ask a hard question to see how RAG is used, I asked a question that has more explicit questions.
- RAG is used! HITL also works correctly, results are sent to the report folder.

Overall, RAG seems to be improving codegen performance. I do think that looking at langfuse we see evidence of it increasing latency, but not by too much, so I do not think it is an issue right now. While it is noticeable as a user that you are waiting, as we continue to improve the model, it should be something to keep in mind.

We can see that it is retrieving relevant information (from FAIS), formatting them so it is readable, and then using that for the codegen. This means the agent can use this context during the codegent, but it does not HAVE to. To improve RAG, one way is to look at scoring, we can let it know what kind of question is being asked, so when scoring happens it is based on the question, not just generally scoring.